Arjun Ghosh

Address
University of Chicago
5801 S Ellis Ave, Chicago
IL 60637, United States

arjunghosh@uchicago.edu
arjunghosh@ieee.org
Personal Webpage
Google Scholar

Employment

Indian Institute of Technology (Bombay), Mumbai, MH, India *Jan 2025 - April 2025*
Project Research Assistant
Working with Dr. Varun Bhalerao (PI: STAR labs)
For space technology development and Cubesat design

University of Moncton, Moncton, NB, Canada *Aug 2023 - Sept 2023*
Research Assistant
Worked in lab of Dr. Viktor Khalack
In Spectropolarimetric data analysis procedures for mCP stars

Education

University of Chicago, Chicago, IL, United States *2025 – 2030 (expected)*
Doctor of Philosophy, Astronomy and Astrophysics
Department of Astronomy and Astrophysics

University of Calcutta, Kolkata, WB, India *June 2020 – May 2024*
Bachelor of Technology (Honors), Electronics and Communication
Institute of Radio Physics and Electronics
Cumulative GPA: 9.26 out of 10.0

Scientific Publications

During Bachelor's

- A. Ghosh and R. Basu Thakur, "A new and flexible design method for Symmetric Quadrature Hybrid Couplers using Markov Chain Monte Carlo," *arXiv preprint* arXiv:2412.09528 (2024). (Journal in preparation)
- A. Ghosh et al., "Solar Observation at 53 MHz using CU-ST Radar with Signal Processing Techniques in Passive Mode Operation," *Springer Book Series, India Radar Meteorology – 2024* (Accepted, to be published).
- A. Ghosh et al., "Low-Frequency Observation of Taurus-A with Signal Processing Techniques in Passive Mode Operation of CU-ST Radar," *Proc. 8th Int. Conf. on Computers and Devices for Communication (CODEC 2023)*, IEEE (2023).

Conferences

- 6th URSI Regional Conference on Radio Science, India; **Oct 2024**
  - My work on APSERa control and data acquisition systems was presented as a project update.

- 6th Conference on Indian Radar Meteorology, India; **Jan 2024**
  - Presented an **oral talk** on "Solar Observation at 53 MHz using CU-ST Radar with Signal Processing Techniques in Passive Mode Operation."
- 8th International Conference on Computers and Devices for Communication, India; **Dec 2023**
  - Presented an **oral talk** on "Low-Frequency Observation of Taurus-A with Signal Processing Techniques in Passive Mode Operation of CU-ST Radar."
- Research Day 2023, Moncton, New Brunswick, Canada; **Sep 2023**
  - Poster presentation on "Study of the Mean Longitudinal Magnetic Field in roAp Star HD99563."
- 41st Meeting of the Astronomical Society of India, Indore, India; Mar 2023
  - Poster presentation on "Solar Observation Using CU-ST Radar in Passive Mode." **Was given the Best Poster Award** in the 'Instrumentation and Techniques' category for 2023 by the **Astronomical Society of India**.
- IEEE Global Microwave and Antenna Conclave, Kolkata, India; Jan 2023
  - Attended as a student participant and volunteer.
- IEEE Microwave Antenna and Propagation Conference, Bangalore, India; Dec 2022
  - Competitively selected for the B.Tech/M.Tech Student Connect Initiative Program as a student invitee.

Proposals

- **Contributed to writing satellite launch proposal to ISRO** (Indian Space Research Organization) for STAR lab, IIT-Bombay.

Honors and Awards

During PhD

- Robert R. McCormick Fellowship, University of Chicago, 2025
- McDonnell Center for the Space Sciences Fellowship, Washington University in St. Louis, 2025 *(declined)*

During Bachelor

- Best Student Paper award (Second Prize), iRAD-2024 (National Conference), Jan 2024
- Scholarship awarded (as a part of the Research Assistantship), University of Moncton, Sep 2023
- MITACS Globalink Research Internship Award (2023), MITACS (Canada), AICTE (India) and SICI (India), Dec 2022
- Best Poster Award, 'Instrumentation and Techniques', 41st Meeting of Astronomical Society of India, Mar 2023
- Joint Winner in Optical Model Demonstration, SPIE-OPTICA-OSI Chapter, University of Calcutta, Aug 2022
- 2nd rank, Lecture Competition, IEEE Photonics Society Chapter-IEEE CUSB, University of Calcutta, May 2022

Computer and Technical skills

- **Programming:** Python, MATLAB, HTML, CSS, JavaScript
- **EM Simulation:** Ansys HFSS, CST Studio Suite
- **CAD:** Autodesk Fusion 360, Ansys SpaceClaim
- **FPGA Development:** Xilinx Vivado
- **Mechanical Simulation:** Ansys Mechanical
- **Analog Simulation:** PSPICE
- **Lab training:** 3D printing (single/dual nozzle), laser cutting, operation in Class 100, 1000, and 10,000 cleanroom environments

Research experience

During PhD

- **Design of Antennas for THz KIDs** *July 2025 onwards*
  Ritoban Basu Thakur, Karthik Ramanathan and QUALIPHIDE Collaboration
  California Institute of Technology, Washington University in St. Louis

  Design and simulation for a THz regime antenna to be tuned for use in Dark Matter detection experiments.

During Bachelor

- **Novel design technique for RF, mm and sub-mm wave devices** *Jan 2024 onwards*
  California Institute of Technology, Pasadena, CA, USA
  Advisor: Dr. Ritoban Basu Thakur

  Research Intern (remote): Developed a novel technique for designing arbitrarily long hybrid couplers with a Markov Chain Monte Carlo-based solver (arXiv:2412.09528). Which allows precise system-level design for couplers and can be easily adapted to a wide range of devices. Used the solver to design and tune hybrid couplers with three, five, seven, and nine sections that outperform literature-reported designs (with potential applications in mm and sub-mm wave astronomy); used the technique to design low-pass and band-pass filters to prove the versatility of the design technique.

- **APSERa Instrumentation to probe Epoch of Recombination** *May 2024 - Nov 2024*
  Raman Research Institute (RRI), Bangalore, KA, India
  Advisor: Dr. Mayuri S. Rao and Keerthipriya Sathish

  Visiting Student (on-site): Developed web apps for APSERa (Radio Telescope, 2–4 GHz) for system monitoring, control, and data acquisition. Stabilized the system at cryogenic temperatures under vacuum through multiple runs, achieving stable temperature runs at 75K, 77K, and 120K. Performed pressure variation studies and leak testing to improve system integrity. Tuned the cryocooler controller by modeling the cryocooler system for temperature stability around setpoints. Designed and integrated a thermal shroud for the cryocooler of APSERa to improve heat dissipation. Conducted antenna and radome simulations to optimize performance and study antenna-environment couplings. Prepared an experimental deployment plan for APSERa. Developed a Python simulation pipeline to analyze RF chain systematics and validated it with MATLAB-CST simulations.

- **Streamlining magnetic field detection of mCP stars** *Sep 2023 – Nov 2023*
  University of Moncton & University of Delaware
  Moncton, Canada & Delaware, United States
  Collaborators: Viktor Khalack (Moncton), Colin Folsom & Christi Erba (Delaware)

  Collaboration (remote): Designed a user interface for operating Python tools such as LSDpy and SpecpolFlow and incorporated them into a single pipeline for easy analysis of spectro-polarimetric data to infer stellar magnetic fields.

- **Calculating magnetic field and rotation period of mCP stars** *Jun 2023 – Sept 2023*
  University of Moncton, Moncton, New Brunswick, Canada
  Advisor: Dr. Viktor Khalack

  Research Intern (MITACS Globalink intern, on-site): Analyzed available spectra of CP stars HD99563 and HD210684, with data from spectropolarimeter ESPaDOnS to calculate their radial velocities and mean longitudinal magnetic fields using techniques like least squares deconvolution. Calculated magnetic field variability to derive stellar rotation periods, which improved the currently known period estimate.

- **Radio Astronomy with CUST-Radar** *Jun 2022 – Jan 2024*
  University of Calcutta & Indian Institute of Technology, Indore
  Kolkata and Indore, India
  Advisors: Dr. Abhirup Datta & Dr. Patra (IIT-I), Dr. Paul (CU)

  Research Student (Partially Funded, On-Site): Characterized the CU-ST radar antenna/system and conducted RFI testing on site to assess suitability for astronomy experiments. Created a Python-based data analysis pipeline for radar spectral data to generate time series plots of experiments performed. Worked on the RADAR to bypass transmissions of the active system and operated in passive mode; performed radio source tracking experiments (Sun, Virgo A, Taurus A) to test the pipeline and calibrate the antenna beam pattern.

Talks and Seminars

- 'MCMC Assisted Coupler Designer Analyzer: A novel technique to design wideband couplers', RRI, India, Nov 2024
- 'Study of magnetic field in roAp star HD99563 using Least Squares Deconvolution', MOBSTER Webinar Dec 2023
- 'Chemical Peculiarities in Stars and analysis of photometric data: An Overview', Ajeenkya D.Y. Patil University, Aug 2023
- 'Our Beautiful Universe: Probes in Space', University of Calcutta, Seminar Dec 2022

Teaching Experience

During Bachelor's

- **Antenna Laboratory (ECHN 2.2.8)**, University of Calcutta, January 2022 - July 2022
  Assisted Dr. Santanu Mondal by taking classes in Computational Electromagnetics and CST simulation lab for students

Leadership and Outreach Experience

Leadership

During Bachelor

- **IEEE Calcutta University Student Branch, University of Calcutta**
  Lead Webmaster, Dec 2022 - Dec 2023
  Awareness and Expansion Team lead, Dec 2022 - Dec 2023
- **IEEE Photonics Society Student Branch Chapter, University of Calcutta**
  Treasurer, Jan 2023 - Jan 2024
- **IEEE Antenna and Propagation Society Student Branch Chapter, University of Calcutta**
  Vice Chair, Jan 2023 - Jan 2024
- **Astronova, Astronomy Club at University of Calcutta, University of Calcutta**
  President and Founder, Dec 2022 - Dec 2023
- **IEEE Microwave Theory and Techniques Society Student Branch Chapter, University of Calcutta**
  Treasurer, Jan 2023 - Jan 2024

Outreach

During Bachelor's

- Designed and launched the official webpage of IEEE CUSB and three associated societies.
- Led awareness drives and talks to encourage student participation in STEM societies.
- Actively participated in outreach events for underrepresented and financially constrained students.
- Organized the "Career Fair with Various Industries," providing students with a platform to connect with industry leaders and gain career insights.
- Led the annual Science and Technology Exhibition, fostering a robust community of STEM professionals.

Professional memberships

- IEEE Student member 2022 - present
- IEEE Antennas and Propagation Society 2022 - present
- IEEE Microwave Theory and Technology Society 2022 - present

References

- ***Dr. Ritoban Basu Thakur***, Staff Scientist Level-3, Jet Propulsions Lab & Visiting Associate, California Institute of Technology, Pasadena, CA, USA
  Email: ritoban@caltech.edu
- ***Dr. Mayuri Sathyanarayan Rao***, Associate Professor, Raman Research Institute
  Email: mayuris@rri.res.in
- ***Dr. Sumitra Mukhopadhyay***, Associate Professor, University of Calcutta
  Email: smrpe@caluniv.ac.in